

to the snake, programmes about the python on Discovery Channel, as well as Python programming.

With such diverse search results, it becomes difficult for new users to make any sense of the results fetched. If what the user really wanted was information on Python programming alone (and not details about the snake), there was no way for the search engine to have deduced this, which was why the search results included a lot of results about the snake. This is why the concept and algorithms of the semantic Web are required.

The semantic Web is focused on understanding search queries on the basis of the syntax, semantics and actual meaning of the keywords entered by the user. The semantic Web assists the user to get the results in a categorised way.

For example, if a user types the keyword 'Python', the semantic Web search engine gives the search results under different categories, as shown below.

**Category 1: Python programming:**

Search result-1, Search result-2, Search result-3, Search result-4.....

**Category 2: Python reptile:**

Search result-1, Search result-2, Search result-3, Search result-4.....

**Category 3: Python movie:**

Search result-1, Search result-2, Search result-3, Search result-4.....

Speech based searches on Google implement semantic technology and speech recognition, whereby the meaning and context of the sentences are understood, based on which responses are given by the Google search engine. But in some cases, speech based searches also give results that may not be accurate.

## Knowledge representation and ontologies

The technology of the semantic Web comprises knowledge representation and ontologies, which extract the

actual meaning of the keywords entered by the user. The knowledge representation techniques used in the semantic Web are such that the database can be associated with the corresponding metadata, and the knowledge can be stored in the databases. The creation of ontology is required so that association in the data objects is possible.

### An example of knowledge associations and ontology

A knows B  
B knows C C is a friend of A B may be knowing D  
D met C  
D is a friend of A

## Limitations of websites without the semantic Web

Without the semantic Web, websites do not have accurate and interrelated information. In general, the base programming language of Web pages is HTML, in which different mark-ups are defined and then the websites are programmed. Yet, there are various limitations and disadvantages with HTML. One of its major limitations is that it is just a mark-up language to render and display Web pages on the Web browser. The text or content written in HTML does not have any relationship. This means that anything can be displayed on the Web browser using HTML, whether it has any association or not.

For example, text like 'A is the spouse of B' and 'B is the son of A' can also be displayed on a Web browser using simple HTML tags. But these types of associations do not occur using knowledge representation techniques in semantic Web based apps.

The main motive of the semantic Web app is to enable the machine to understand the content and its relationships, which are defined in the Web pages. Using semantic Web based knowledge representation, direct queries can be asked from the search engines like, "What is

my date of birth?", "Which is my birth place?" or any other. Using the semantic Web, the ontologies will scan the available information about the specific person from multiple sources and then give accurate results. In this manner, a Web based expert system can be developed using semantic Web technology.

There are different types of standards associated with knowledge representation and ontologies so that data definition and interrelationships are created.

The following are the data formats and standards of the semantic Web.

- **XML:** eXtensible Markup Language
- **RDF:** Resource Description Framework
- **OWL:** Web Ontology Language
- **FOAF:** Friend Of A Friend
- **SPARQL:** Recursively known as SPARQL Protocol and RDF Query Language

## Friend Of A Friend (FOAF)

This standard defines the ontology in machine readable format so that the description of different objects and their relationships can be specified.

For example, as in social media portals, the relationships are defined and extracted by the algorithms. In Facebook, Twitter and Instagram, these associations are grouped and then predicted. Generally, in social media, users receive suggestions like, "You may know these persons," or "You may be interested in following these people," and so on. These types of relationships are extracted using FOAF profiles.

Whenever new information is uploaded by the user on social media, the relationships are built using ontology and FOAF. After that, people with the matching profile, and who have met the person at a specific place are extracted. And such people are the ones suggested to users — "You may know these people."